

Open Source Contribution – Pathfinding Algorithms Repository

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Project Overview

Contributed documentation and repository management files to a Python-based Pathfinding and Graph Search Algorithms visualization project developed using PyGame.

The repository focuses on implementing and visualizing multiple pathfinding algorithms including A, *BFS*, *DFS*, *Dijkstra*, *Theta*, Bellman-Ford, and many more.

Contributions Made

1. CONTRIBUTING.md

Created a contributor guideline document to help new developers understand:

- Repository setup
 - Contribution workflow
 - Coding practices
 - Pull request process
 - Contribution ideas
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2. CODE_OF_CONDUCT.md

Added a professional community guideline document to maintain:

- Respectful collaboration
 - Positive communication
 - Open-source professionalism
 - Contributor safety and inclusivity
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3. PULL_REQUEST_TEMPLATE.md

Designed a pull request template to standardize:

- Change descriptions
 - Testing details
 - Documentation updates
 - Contributor checklists
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4. ISSUE_TEMPLATE.md

Created a structured issue reporting template for:

- Bug reporting
 - Reproduction steps
 - Expected behavior
 - System information
 - Debugging support
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Technologies Used

- Python
 - PyGame
 - Git
 - GitHub
 - Markdown Documentation
 - Open Source Workflow
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Skills Demonstrated

- Open Source Contribution
 - GitHub Repository Management
 - Technical Documentation
 - Collaboration Workflow
 - Software Development Practices
 - Version Control
 - Community Contribution Standards
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Learning Outcome

This contribution helped improve understanding of:

- Real-world open-source collaboration
 - GitHub pull request workflow
 - Repository structure management
 - Documentation standards
 - Professional software contribution practices
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Repository Domain

Pathfinding Algorithms and Graph Search Visualization

Topics involved:

- Artificial Intelligence
 - Graph Theory
 - Algorithm Visualization
 - Search Algorithms
 - Maze Generation
 - Data Structures
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Contribution Summary

Successfully prepared and contributed professional repository support files to enhance onboarding, collaboration, and contribution workflow for future developers in the open-source project.

Platforms Used

- GitHub
 - Visual Studio Code
 - Python Environment
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Career Relevance

This contribution demonstrates practical exposure to:

- Open-source ecosystem
 - Software collaboration standards
 - Technical communication
 - Repository documentation practices
 - Professional developer workflows
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Prepared By

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